

Lecture Summary

Introduction: Extending Beyond ARIMA

This lecture builds on ASQ-01 by moving from mean-return modeling to volatility modeling and cost-aware strategy design.

It shows why assuming constant volatility is unrealistic and how ARCH/GARCH models provide more accurate forecasts for risk and position sizing.

1. Parameter Selection & Automation

- We performed manual ARIMA order selection on PYPL weekly data in ASQ-01.
- In this lecture, we'll explore an automated approach to apply the same technique to various assets, such as SPY, QQQ, IWM, and TSLA, using the `sktime` package.
- The automation yielded an improvement for only one asset, and underperformed on the other three:

Comparison of automated vs manual parameter selection:				
Asset	Manual(2,1,2)	sktime	Selected Order	
SPY	-11.2%	-51.1%	(0, 1, 0)	-
QQQ	-6.4%	-54.6%	(0, 1, 0)	-
IWM	-14.7%	-67.7%	(1, 0, 2)	-
TSLA	-12.7%	+23.8%	(1, 2, 2)	+

It suggests data limitations and also that automation doesn't necessarily lead to superior results.

2. Transaction Costs in Strategy Evaluation

- The performances we obtained in the previous step didn't consider transaction costs.
- In this step, we assumed a transaction cost of 0.03% per trade.

- Here are the results:

Performance with transaction costs:

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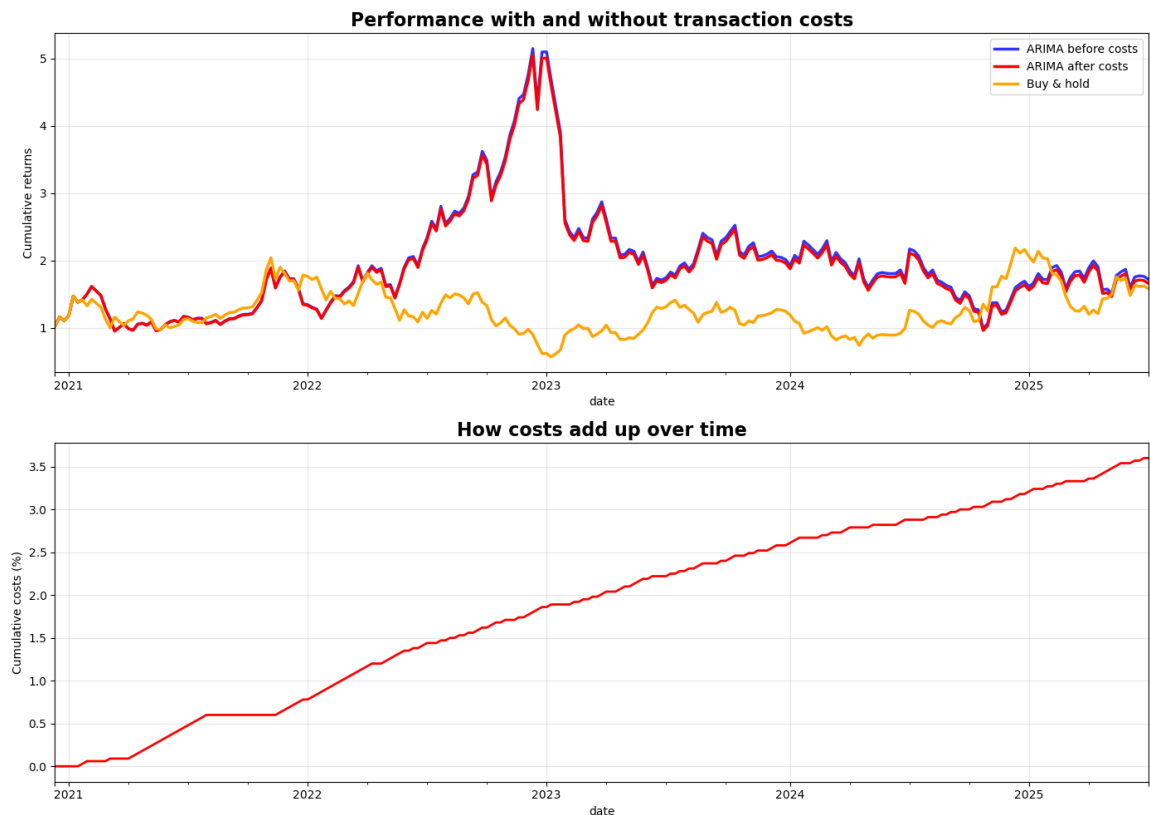
Before costs: 71.8%
 After costs: 65.7%
 Buy & hold: 57.9%
 Cost impact: 6.0%
 Total trades: 120
 Note: Using US market transaction costs (0.03% per trade)
 PYPL trades on NASDAQ with tight spreads and zero commissions

Excess returns:

Before costs: 13.8%
 After costs: 7.8%

Strategy maintains positive excess returns after costs.
 Cost impact: 44% reduction in excess returns

- Here are visual representations of the impact of transaction costs:



- Thus, transaction costs eat into our excess profits and accumulate considerably over time. When backtesting, it's necessary to consider these costs with realistic values so that the backtest results are realistic.

3. Time-Varying Volatility

- Till the previous step, we implicitly treated volatility as constant.
- However, it's an observed fact in the markets that volatility exhibits clustering.
- These are the different volatility regimes for the PYPL stock:

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Examining PYPL volatility over time...
PYPL volatility by period:
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Pre-COVID (2019)      : 47.7% avg (29.2% - 69.1% range)
COVID period (2020)   : 84.1% avg (33.2% - 139.4% range)
2021 recovery         : 51.6% avg (22.2% - 90.3% range)
2022 bear market      : 64.3% avg (36.4% - 87.8% range)
Recent (2023-2025)    : 60.8% avg (32.1% - 114.0% range)
```

- Here's the visual representation of the same:



- One approach would be to set the risk exposure based on the current volatility regime.
- Another approach is to adjust the position size based on the volatility forecast.
- For these forecasts, we use the ARH and GARCH models.

4. ARCH Models for Volatility Forecasting

- We implemented an ARCH (1) model:

ARCH(1) model:
 $\sigma^2(t) = \omega + \alpha \times \varepsilon^2(t-1)$

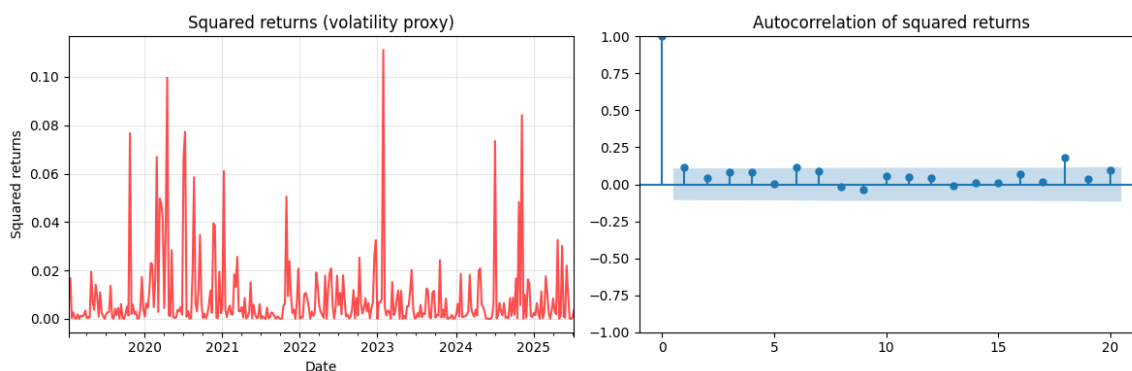
Where:

- $\sigma^2(t)$ = today's variance
- ω = baseline variance level
- α = impact of yesterday's shock
- $\varepsilon^2(t-1)$ = yesterday's squared returns

- The results are:

ARCH test results:
 Test statistic: 8.186
 p-value: 0.1463
 Limited ARCH effects found

- Here's a visual representation:



- Volatility clustering is confirmed if the bars in the autocorrelation plot extend beyond the blue band.
- Next, we used the rolling ARCH (1). Here are the results:

ARCH model insights:

- Volatility forecasts: 9.77% average (7.18% - 17.31% range)
- Alpha parameters vary from 0.2897 to 0.6763
- Models detect modest volatility clustering
- Ready for position sizing application

- The ARCH model doesn't capture long-term persistence of volatility, as seen from the position sizing below:

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ARCH position sizing (last 10 periods):
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Date      Vol forecast  Position size  Weekly risk
-----
2021-09-19    8.78%    $ 22,770    $ 2,000
2021-09-26    7.42%    $ 26,963    $ 2,000
2021-10-03    7.30%    $ 27,404    $ 2,000
2021-10-10    7.44%    $ 26,884    $ 2,000
2021-10-17    7.18%    $ 27,855    $ 2,000
2021-10-24    8.10%    $ 24,677    $ 2,000
2021-10-31    8.29%    $ 24,123    $ 2,000
2021-11-07   17.31%    $ 11,553    $ 2,000
2021-11-14    9.27%    $ 21,585    $ 2,000
2021-11-21   16.21%    $ 12,341    $ 2,000

Position size range: $11,553 - $27,855
Variation ratio: 2.4x
Risk per trade stays around $2,000

```

- We overcome this by using the GARCH model.

5. GARCH Models and Extensions

- The GARCH model is an extension of the ARCH model, and captures volatility persistence:

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ARCH(1) model:
 $\sigma^2(t) = \omega + \alpha \times \varepsilon^2(t-1)$ 

GARCH(1,1) model:
 $\sigma^2(t) = \omega + \alpha \times \varepsilon^2(t-1) + \beta \times \sigma^2(t-1)$ 
*****
New: volatility persistence

Where:
-  $\alpha$  = impact of recent shocks (like ARCH)
-  $\beta$  = persistence of recent volatility (new)
-  $\alpha + \beta$  = total persistence (should be < 1)

Higher  $\beta$  means volatility shocks last longer

```

- In this example, the GARCH (1,1) model doesn't significantly outperform the ARCH (1) model:

GARCH(1,1) vs ARCH(1) comparison...

Forecast accuracy comparison:

Model	Mean Error	Correlation
ARCH(1)	5.678	0.152
GARCH(1,1)	5.610	0.121

GARCH improvement: 1.2% lower forecast error

Mixed results - GARCH doesn't clearly outperform ARCH

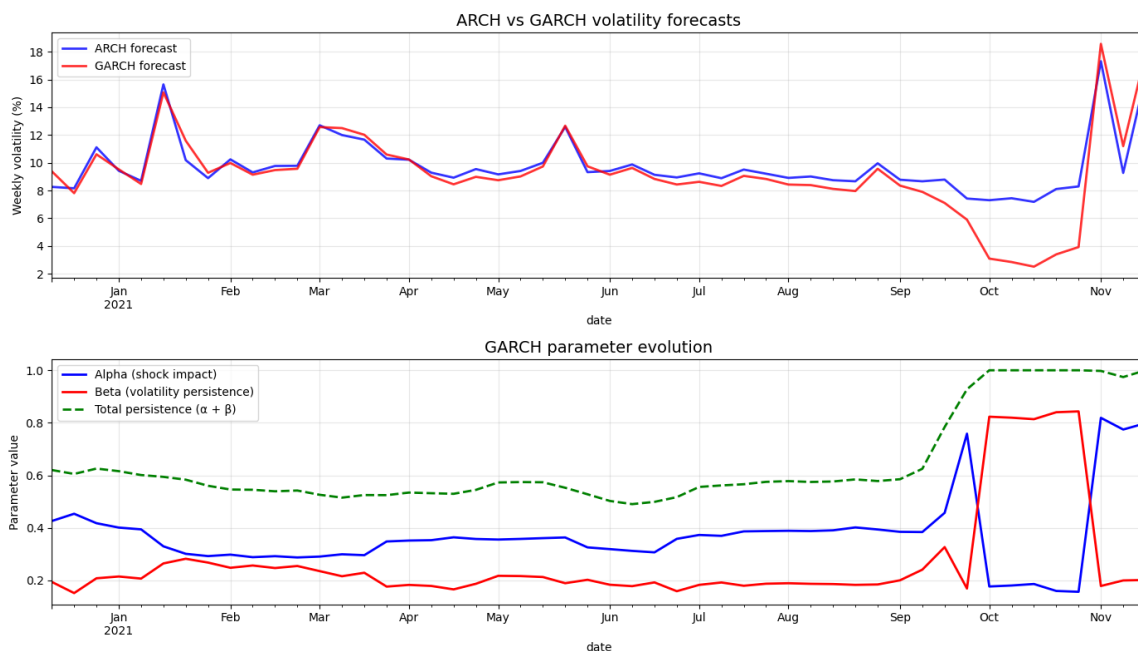
Parameter evolution:

Alpha variation: 0.6624

Beta variation: 0.6918

Parameters vary significantly over time - models adapt to changing conditions

- However, the characteristic of the GARCH is that the model adapts to changing conditions over time:



- The GARCH model also adjusts the position based on volatility more dynamically:

GARCH-based dynamic position sizing...

GARCH position sizing (last 10 periods):

```
=====
```

Date	Vol Forecast	Position Size	Weekly Risk	Persistence
-----	-----	-----	-----	-----
2021-09-19	7.10%	\$ 28,170	\$ 2,000	0.784
2021-09-26	5.90%	\$ 33,919	\$ 2,000	0.928
2021-10-03	3.09%	\$ 64,711	\$ 2,000	1.000
2021-10-10	2.84%	\$ 70,340	\$ 2,000	1.000
2021-10-17	2.51%	\$ 79,669	\$ 2,000	1.000
2021-10-24	3.40%	\$ 58,908	\$ 2,000	1.000
2021-10-31	3.92%	\$ 51,011	\$ 2,000	1.000
2021-11-07	18.56%	\$ 10,774	\$ 2,000	0.998
2021-11-14	11.19%	\$ 17,865	\$ 2,000	0.974
2021-11-21	17.86%	\$ 11,201	\$ 2,000	1.000

Position sizing insights:

Position range: \$10,774 - \$79,669

Size variation: 7.4x

Risk per trade: ~\$2,000 (constant)

Significant position adjustment based on volatility

- At this stage, we have ARIMA models for predicting price direction and GARCH models for estimating volatility. In the next step, we'll combine both.

6. Integrated Modeling Approaches

- In this step, we integrate the rolling ARIMA model with the rolling GARCH model.
- Here's the performance of the rolling ARIMA-GARCH integrated modeling:

Integrated ARIMA-GARCH: +65.7%

Buy and hold: +57.9%

Excess return: +7.8%

Trading activity:

Total trades: 120

Long signals: 135 (56%)

Short signals: 104 (44%)

Integrated approach shows positive excess returns

GARCH volatility forecasts help with position sizing

7. Implementation & Practical Considerations

- Checklist for deployment:
 - Data pipeline
 - Model automation
 - Risk management
 - Execution system
 - Monitoring
- Guidelines for implementation:
 - Start simple
 - Monitor everything
 - Plan for failure
 - Risk first, returns second
- Next steps:
 - Immediate
 - Medium term
 - Long term

Summary Insights

- ARIMA models capture direction but miss volatility
- GARCH models improve understanding of risk and clustering
- Transaction costs often turn profitable backtests into losing strategies
- Integration of mean and volatility forecasts is a good approach
- Robust, adaptive frameworks are required for actual trading in the markets